Theodore Nguyen

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Education

University of Minnesota – Twin Cities Minneapolis, MN
Master of Science in Computer Science *Expected May 2027*
Bachelor of Science in Computer Science *May 2025*

Skills

Languages: Python; Java; C; C++; C#; CSS/HTML/TS/JS; SQL; OCaml
Tools: AWS (S3, Aurora, Bedrock); Azure (DevOps, IoT); Docker; Git; GitLab; Jira; Linux; MySQL; PostgreSQL; Postman; MatLab; Unity; Visual Studio; VSCode

Work Experience

Medtronic Newton, MA
Software Engineering Intern *May 2026 – August 2026*

- Engineered a data post-processing pipeline to extract, merge, and transform .h5 and .case files from AWS S3 into CSVs, successfully processing 2,000+ distinct cases for scalable storage in an Amazon Aurora database
- Developed a natural language text-to-SQL chatbot utilizing **AWS Bedrock** and **LangGraph** to directly query the 2,000+ case dataset, implementing Pydantic for schema validation and SQLAlchemy for dynamic query construction
- Built an interactive React frontend leveraging Apache ECharts and ECharts-GL to automatically render customized 2D and 3D data visualizations directly from user chat prompts

Daikin Applied Americas Plymouth, MN
Software Engineering Intern (Cloud) *May 2025 – April 2026*

- Developed and published **NuGet packages** for shared components, reducing new project setup time by ~15–20% and standardizing internal code reuse
- Designed and implemented RESTful APIs in **C#** and with SQL using an API-first approach with Swagger, improving cross-team collaboration and reducing integration issues. Utilized **Postman** for integration testing

Bracco Medical Technologies Eden Prairie, MN
Software Engineering Intern *May 2024 – August 2024*

- Designed and developed a client-facing UI using **Angular** and **React** to display cardiovascular injector data as Key Performance Indicators, leveraging **Redash** and **PostgreSQL**. Integrated **REST API** calls using FHIR
- Deployed both the web application and FHIR server as **Azure IoT Edge** modules, allowing containers to be deployed to downstream devices
- Modified **Azure Pipelines** file to include automated unit testing and linting of the web application which decreased deployment time by **40%** based on testing

Earl E. Bakken Medical Devices Center Minneapolis, MN
3D Printing and Segmentation Specialist *June 2023 – Present*

- Create and upload 3D models to VR using **Unity** to allow clinicians to view and interact with anatomical models
- Work with 50+ doctors and students to design and prototype their medical inventions and desired anatomical models
- Create 3D models of patient anatomy using CT scans in Mimics to 3D print and help visualize patients' physical condition

Undergraduate Research Assistant *June 2023 – October 2024*

- Developed a mixed reality guided nasogastric tube placement program using Unity and **OpenPose**, enabling clinicians to visualize tube positioning within the patient's body accurately
- Used Unity to develop a VR surgical trainer for a trochleoplasty surgery that incorporates 3D anatomical models and **C#** coding to apply scripts to in game objects that allows surgeons to train for a surgery without entering the operating room

Projects

Local Multimodal RAG System

- Developed a local Retrieval-Augmented Generation (RAG) system to query provided documents. Utilized unstructured for multimodal PDF parsing, Llama-Index and ChromaDB for vector indexing, and Ollama to run local models

Extracurricular

Society Asian Student Engineers | Labs Co-Director *October 2023 – May 2026*

- Work and mentor a team of students to design and engineer a project every year (ex. drone, mechanical heart)